



Project Part 2: Robust Head Pose Estimation Under Facial Occlusions

Antonio Vila Leis & Enric Baixauli Casañ

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1 Problem statement

This project is based on the domain of robust Head Pose Estimation (HPE). Through the analysis of recent scientific literature and surveys like R. Neupane et al. (2024)[1] or L. Zhou et al. (2023)[2], we identified that although 3D Human Pose Estimation (3D-HPE) has advanced significantly thanks to deep learning, occlusion remains one of the most critical challenges. In real-world scenarios, occlusions caused by external objects (such as glasses, surgical masks or hair) degrade the performance of these models.

One of the main reasons is that the majority of training datasets are captured in controlled environments, lacking the necessary diversity for "in-the-wild" scenarios. This lack of diversity causes models to exhibit poor generalization when faced with samples outside their distribution, such as partially covered faces.

To address this challenge, we follow the guidelines of R. Neupane et al. (2024)[1], who, in their section on "Occlusion Handling", propose two main research directions: the use of diffusion-based models to reconstruct occluded body parts or the application of Data Augmentation techniques to train models with datasets with real or simulate occlusions. We chose the second option, so our research focuses on investigating whether a modern Transformer-based architecture can overcome this bias through customized Data Augmentation.

Solving Head Pose Estimation (HPE) under occlusion is critical for the successful real-world deployment of systems in fields such as driver attention monitoring to prevent accidents or Human-Computer Interaction (HCI). Given that the use of surgical masks and sunglasses is currently more common than ever, a model that collapses in the face of these occlusions becomes useless and potentially dangerous in production environments.

Following the required formal structure, our research question is: "Can a Vision Transformer-based architecture address the challenge of performance degradation caused by facial occlusions, such as glasses and masks, on images of real human faces?"

Our Initial Hypothesis is that standard Vision Transformer architectures, when pre-trained on datasets without occlusions (such as 300W_LP or AFLW2000-3D [3]), implicitly assume a clean facial topology, which causes a degradation in the quality of their results in our field. By applying a Data Augmentation pipeline that simulates these occlusions, we expect that the Transformer's self-attention mechanism will learn to reassign its weights towards unoccluded secondary facial features, thereby mitigating the error.

2 Related work

This section reviews relevant literature across three primary domains: landmark-free HPE, occlusion-handling methodologies and Vision Transformers in spatial reasoning.

Landmark-Based vs. Landmark-Free HPE

Early HPE methods relied heavily on facial landmark detection to solve the 2D-to-3D pose problem via Perspective-n-Point (PnP) algorithms. While effective in controlled settings, landmark-dependent models exhibit catastrophic failure modes under occlusion, as missing or covered keypoints lead to severe geometric distortion.

To overcome this vulnerability, modern research has shifted toward landmark-free, direct regression approaches. State-of-the-art models typically treat HPE as a multi-loss task, utilizing combination classification-regression networks to predict continuous yaw, pitch and roll angles directly from raw pixel intensity values. While convolutional networks (CNNs) like WHENet[4] and SixDRepNet[5] adopt this approach successfully, they have their own limitations that we will mention in the architecture selection section.

Occlusion Handling and Data Augmentation

To mitigate the performance drop caused by objects like masks, sunglasses or hands, the literature suggests two main paradigms:

- Generative Reconstruction uses architectures such as Generative Adversarial Networks (GANs) or Diffusion Models to dynamically inpaint or de-occlude the face before estimating the pose. However,

these pipelines are computationally costly, making them impractical for real-time applications such as driver monitoring.

- The second paradigm focuses on Data Augmentation and Robust Training by exposing the network to synthetic or real occlusions during training. Standard techniques like Random Erasing or CutOut simulate missing data but lack semantic realism. To address this, recent studies such as FSErasing [6] introduced face-parsing augmentation pipelines that mask specific facial regions based on semantic labels rather than random shapes. Their findings demonstrated that forcing a network to train on semantically structured occlusions significantly improves robustness against real-world distribution shifts.

Vision Transformers (ViTs) in Head Pose Estimation

The introduction of the Vision Transformer (ViT) has shifted the backbone paradigm away from Convolutional Neural Networks (CNNs). While CNNs rely on localized receptive fields, ViTs utilize a global self-attention mechanism to model long-range spatial dependencies across the entire image.

This architectural strength is explored in TokenHPE [7], which employs a Transformer structure with specialized orientation tokens to map “critical minority relationships” and facial changes. TokenHPE demonstrates that a Transformer can learn deep cross-orientation relationships even under facial occlusions or low illumination.

Positioning of Our Work

Our work consists of both semantic data augmentation and transformer adaptation. Unlike exploratory studies that merely propose new architectures, we use TokenHPE as our baseline and evaluate its capacity to adapt to feature occlusions. By developing a landmark-bound occlusion pipeline (generating Mask, Glasses and Both variations) and applying it to the previously unseen I2Head dataset [8], we explicitly force TokenHPE’s internal self-attention mechanism to adapt its weights.

3 Architecture selection

Choosing the right backbone architecture is crucial for ensuring model robustness against anomalies, requiring theoretical arguments and results that validate them. We wanted to try three different models: WHENet, SixDRepNet and TokenHPE, all three of them trained on the same HPE datasets.

From a structural point of view, traditional Convolutional Neural Network (CNN) frameworks (such as those in WHENet and SixDRepNet) rely on localized receptive fields and rigid spatial hierarchies. While highly efficient on clean datasets, these convolutional models expect a predictable facial geometry, making them vulnerable to domain shifts when physical occlusions disrupt local pixel relationships.

On the other hand, TokenHPE uses a Vision Transformer (ViT) architecture that treats head pose estimation as a dynamic token-interaction problem. By dividing the image into spatial patches and processing them globally via a self-attention mechanism, TokenHPE can model dependencies across the entire face. If a primary feature like the mouth or nose is obscured, its specialized orientation tokens can theoretically reallocate attention weights to unoccluded regions, such as the forehead or hairline, to calculate an accurate continuous rotation matrix.

To verify this theoretical advantage before initiating any fine-tuning, we executed a baseline test located in *model_demo.ipynb*. We loaded the three models with their official pretrained weights and tested them on the same image, a randomly selected image from the I2Head dataset. The resulting execution output was the following:

Ground Truth	Pitch	Yaw	Roll	
	+2.75°	-1.98°	+17.35°	
Model Identifier	Pitch (Error)	Yaw (Error)	Roll (Error)	MAE
TokenHPE	+6.98 (4.23°)	+6.46 (8.45°)	+11.93 (5.42°)	6.03°
SixDRepNet	-9.16 (11.91°)	+3.85 (5.84°)	+2.13 (15.22°)	10.99°
WHENet	-16.07 (18.82°)	-1.43 (0.55°)	-5.47 (22.82°)	14.06°

Table 1: Model predictions, angular errors and MAE for a sample with known ground truth.

Just as we expected, these results validated our hypothesis, the Mean Average Error (MAE) of the TokenHPE was almost half of the next best model SixDRepNet, while WHENet delivered the poorest results.

4 Experimental setup

Dataset and Occlusion System

Initially, our approach considered performing the fine-tuning and evaluation of the model using the 300W_LP dataset. To simulate facial occlusions, we developed a custom system that takes as input the 68 facial keypoints provided by the .mat files of that dataset, dynamically drawing polygons over the ocular area to simulate glasses and over the lower region (jaw, mouth and nose) to simulate masks.

However, after deeply analyzing the base model (TokenHPE), we realized that it had already been pre-trained on the main state-of-the-art datasets, including 300W_LP. Evaluating our hypothesis on this same dataset would have introduced a severe bias due to data leakage, invalidating the evaluation and the real generalization capability. Therefore, we discarded 300W_LP for the testing phase and focused on finding a completely new domain.

Finally, we selected the I2Head Database (created by the Public University of Navarre, UPNA) [8]. Although this dataset provided high-quality images with precise pose labels, it lacked the 68 landmarks necessary to run our occlusion system. To solve this problem, we used FaceAlignment [9], a library based on deep neural networks (Face Alignment Network or FAN architecture). This tool allowed us to detect and extract with high precision the spatial coordinates of the 68 structural facial points directly from the raw images. In this way, we were able to automatically generate our own synthetic dataset from I2Head. The images were divided into four modalities: clean faces (Raw), with glasses (Glasses), with a mask (Mask) and with both occlusions simultaneously (Both).

Evaluation Protocol and Metrics

To ensure the validity of the results and measure the generalization of the architecture, we applied a data partition based on the subjects' identities, avoiding having images of the same person in different sets. The I2Head database has a total of 12 individuals, so we used 9 subjects (738 images) for train, 1 subject (82 images) for validation and 2 subjects (164 images for each of the 4 occlusion modalities) for test.

During the optimization phase, we employed the Mean Absolute Error (MAE) as the loss function. For the quantitative evaluation of the Euler angles (Yaw, Pitch, Roll) on the Test set, we used three complementary metrics:

- **MAE (Mean Absolute Error):** Measures the average magnitude of the errors in the predictions, maintaining the original units (radians).

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

- **RMSE (Root Mean Squared Error):** Calculates the square root of the mean squared error. This metric squares the errors before averaging them, which penalizes errors of large magnitude more heavily.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

- **NRMSE (Normalized Root Mean Squared Error):** It is a scale-invariant metric. It is calculated by normalizing the RMSE by the range of the actual distribution ($y_{max} - y_{min}$).

$$NRMSE = \frac{RMSE}{y_{max} - y_{min}}$$

Preprocessing and Fine-Tuning

In the preprocessing stage, all images were resized to the native resolution required by TokenHPE (224×224 pixels) and converted into normalized tensors. To adapt the network to the new domain, we employed two transfer learning strategies: LoRA (Low-Rank Adaptation)[10] and Full Fine-Tuning.

5 Results

Baseline Performance

The quantitative evaluation of the base model shows acceptable performance on unoccluded faces (Raw), achieving a total MAE of 7.95° and an RMSE of 9.92° . However, as initially hypothesized, the model’s performance degrades when faced with occlusions. The most notable deterioration occurs in the Mask mode, where the RMSE jumps to 14.45 and the NRMSE reaches 0.804. This quantitatively demonstrates that the base architecture is highly dependent on the lower facial topology for orientation inference.

Full Fine-Tuning Adaptability

As can be seen in the results, the fully fine-tuned model adapts perfectly to the new dataset, reducing the overall error in the Raw test set (from a baseline RMSE of 9.92° to 6.82°). But the most relevant point here is that the error reduction is proportionally much greater in the occluded areas compared to the raw data. For instance, in the Mask modality, the baseline RMSE of 14.45 $^\circ$ was reduced to only 6.69° . The model stabilizes its error at around 6.7° RMSE across all modalities (Raw, Mask, Glasses, Both). This shows that the model is not only adapting to the new dataset but has successfully learned to ignore severe occlusions, achieving occlusion-invariant predictions.

LoRA vs. Full Fine-Tuning

As part of our experiments, we also evaluated Low-Rank Adaptation (LoRA) [10] as an efficient transfer learning alternative. It proved to be a computationally superior technique, freezing a large part of the model and reducing training times. While LoRA demonstrates competitive performance, it struggles with heavier occlusions. Specifically, Full Fine-Tuning provides strictly better results in the most challenging scenarios, outperforming LoRA in both the Mask (RMSE 6.69° vs. 8.58°) and Both (RMSE 6.69° vs. 8.26°) categories. Consequently, since both models were trained, we discarded the LoRA approach and selected Full Fine-Tuning as our most robust model.

Euler Angles Analysis

An independent analysis of the Euler angles (Yaw, Pitch, Roll) reveals that the Yaw axis (lateral rotation) consistently produces the worst results across all tested models. Even in our Full Fine-Tuning model, the Yaw MAE stands at 9.9° , compared to 5.1° for Roll and a highly accurate 1.5° for Pitch. This indicates that predicting the exact degree of lateral head turn is the most complex challenge in our case.

Qualitative Analysis

To complement the quantitative metrics, Figure 1 illustrates the 3D axis projections plotted directly onto the test subjects. These visual results confirm the numerical findings, demonstrating how the Full Fine-Tuned model better maintains correct axis alignment even when the face is heavily occluded by masks and glasses.

Table 2: Unified Quantitative Results. Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) are reported in degrees ($^{\circ}$), while NRMSE is a scale-invariant ratio. Lower values indicate better performance.

Model	Modality	MAE by Axis ($^{\circ}$)			Total Metrics		
		Yaw	Pitch	Roll	MAE ($^{\circ}$)	RMSE ($^{\circ}$)	NRMSE
Baseline	RAW	7.02	9.00	7.84	7.95	9.92	0.678
	MASK	22.54	8.98	4.70	12.08	14.45	0.804
	GLASSES	10.01	8.43	8.06	8.83	10.56	0.667
	BOTH	20.28	9.62	4.05	11.32	13.59	0.770
Full Fine-Tuning	RAW	10.03	1.58	5.10	5.57	6.82	0.291
	MASK	9.89	1.47	5.04	5.47	6.69	0.284
	GLASSES	9.87	1.58	5.11	5.52	6.75	0.289
	BOTH	9.83	1.48	5.08	5.47	6.69	0.284
LoRA	RAW	7.86	4.01	4.76	5.54	6.90	0.373
	MASK	10.75	4.80	6.25	7.26	8.58	0.449
	GLASSES	7.10	3.41	5.55	5.36	6.64	0.348
	BOTH	9.79	4.56	6.78	7.05	8.26	0.436

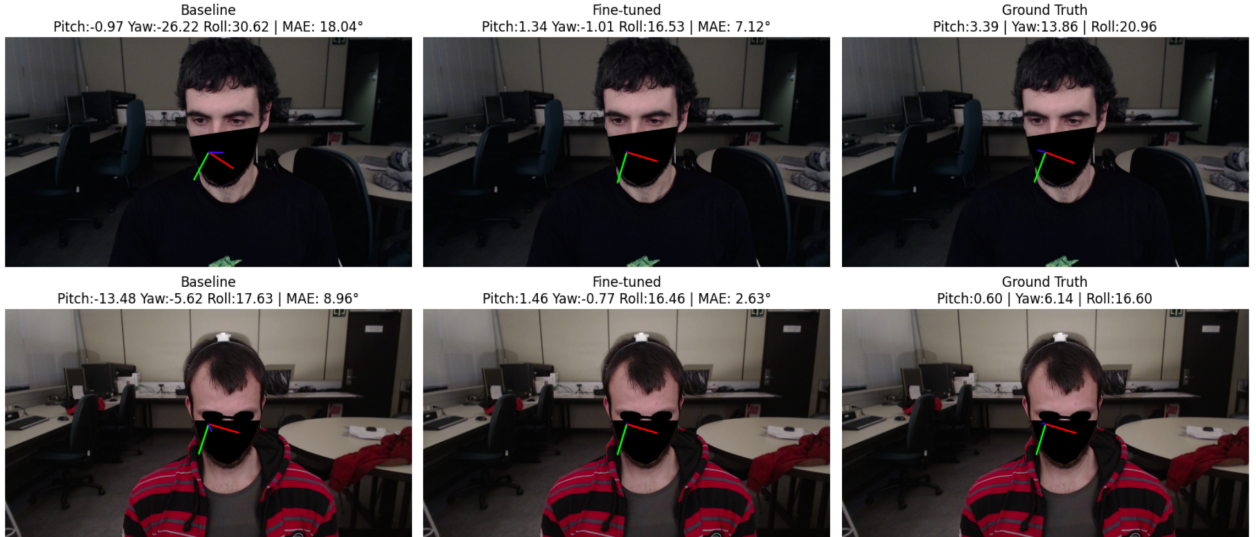


Figure 1: Qualitative analysis and 3D head pose estimations predicted by the Full Fine-Tuned model.

6 Failure analysis

While our fine-tuned TokenHPE exhibits strong overall resilience compared to traditional convolutional networks, exploring its limits is essential to understand its behaviour. To evaluate the model’s vulnerabilities, we subjected the architecture to two distinct failure scenarios designed to test its geometric resolution boundaries and its behavior when encountering unknown spatial features.

The first failure mode examines the impact of the scale and distance of the target. To evaluate this, we passed an identical image of a subject at a fixed pose through three decreasing scales (Base, Small and Smaller scale), testing the progression across both a complex office background and a fully white one.

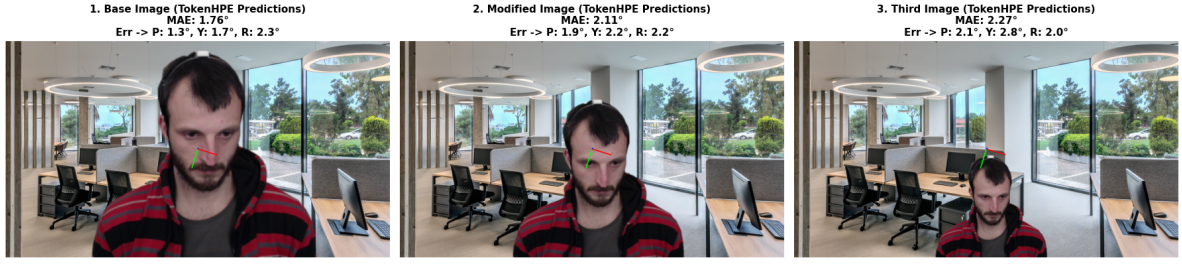


Figure 2: Scenario A visual results

Image Version	Pitch (Error)	Yaw (Error)	Roll (Error)	MAE
Ground Truth	+0.36°	+0.10°	+14.73°	–
1. Base Image	+1.61 (1.26°)	–1.61 (1.71°)	+17.06 (2.33°)	1.76°
2. Small Image	+2.26 (1.91°)	–2.10 (2.20°)	+16.96 (2.23°)	2.11°
3. Tiny Image	+2.43 (2.08°)	–2.66 (2.76°)	+16.71 (1.97°)	2.27°

Table 3: Scenario A: Office Background, Fixed Pose, Raw Environment (No Occlusion)

Image Version	Pitch (Error)	Yaw (Error)	Roll (Error)	MAE
Ground Truth	–1.04°	+0.65°	+14.08°	–
1. Base Image	+1.12 (2.17°)	–1.28 (1.92°)	+16.12 (2.05°)	2.05°
2. Small Image	+1.73 (2.77°)	–1.23 (1.87°)	+16.16 (2.08°)	2.24°
3. Tiny Image	+2.34 (3.38°)	–1.03 (1.68°)	+16.67 (2.59°)	2.55°

Table 4: Scenario B: White Background, Fixed Pose, Compound Environment (Glasses & Mask)

The resulting metrics show a clear trend: as the physical scale of the face decreases, the Mean Absolute Error (MAE) progressively increases across both evaluation environments. This is directly tied to the tokenization process of the Vision Transformers. When the face is large, the ViT distributes key facial features across multiple tokens, allowing the self-attention mechanism to accurately model their relationship. However, when the face is small, these features collapse within a smaller number of tokens, depriving the network of the necessary spatial resolution and degrading the pose estimation.

The second failure mode evaluates the model’s response to unseen and out-of-distribution occlusions. Because our training and optimization pipeline exposed the network exclusively to specific overlays (masks and glasses), we introduced completely new obstructions (large over-ear headphones and a studio microphone) to evaluate the model’s capacity to generalize.

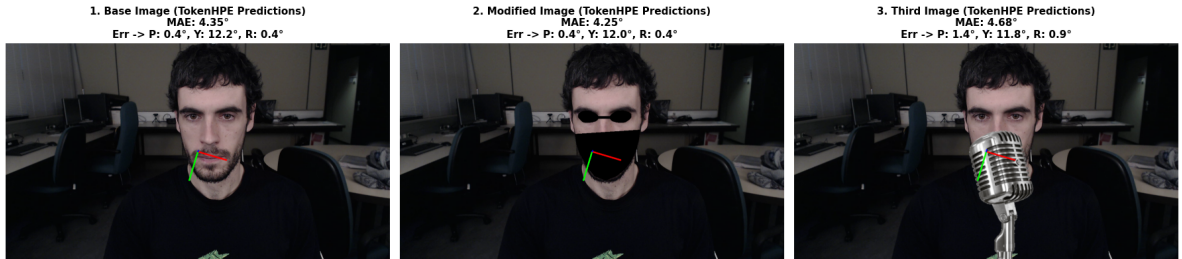


Figure 3: Scenario D visual results

Image Version	Pitch (Error)	Yaw (Error)	Roll (Error)	MAE
Ground Truth	+0.36°	+0.10°	+14.73°	–
1. Base (Raw)	+1.32 (0.96°)	–1.46 (1.56°)	+16.56 (1.83°)	1.45°
2. Mod (Glasses)	+1.34 (0.98°)	–1.15 (1.25°)	+16.38 (1.65°)	1.29°
3. Third (Phones)	+1.61 (1.25°)	–1.71 (1.81°)	+16.76 (2.02°)	1.70°

Table 5: Scenario C: Standard Background, Fixed Pose (Unseen Occlusion: Headphones)

Image Version	Pitch (Error)	Yaw (Error)	Roll (Error)	MAE
Ground Truth	+1.58°	+10.75°	+15.85°	–
1. Base (Raw)	+1.15 (0.43°)	–1.42 (12.17°)	+16.30 (0.45°)	4.35°
2. Mod (Both)	+1.22 (0.37°)	–1.22 (11.97°)	+16.27 (0.41°)	4.25°
3. Third (Mic)	+2.98 (1.40°)	–1.00 (11.75°)	+16.75 (0.90°)	4.68°

Table 6: Scenario D: Standard Background, Fixed Pose (Unseen Occlusion: Microphone)

As expected, introducing these new occlusions caused a distinct degradation in performance, with the large headphones jumping to an MAE of 1.70° and the microphone pushing errors to 4.68°. This confirms that the model’s learned occlusion invariance is tied to the specific structures it encountered during training, meaning it treats unfamiliar objects as corruptions rather than adapting around them.

When looking at the results, there’s a surprising pattern: the model sometimes performs better on faces with occlusions than on the original, unobstructed faces. For example, in Scenario C the glasses version reaches an MAE of 1.29°, surpassing the raw face at 1.45°. In Scenario D, the mask-and-glasses model gets 4.25°, slightly better than the raw input’s 4.35°. This happens because fine-tuning reshapes the model’s internal expectations. While training on masked and glasses-covered faces, the network learns to rely on alternative geometric cues like edges, textures, and the specific unoccluded regions that remain visible. Its attention mechanism becomes specialized for those patterns.

As a result, when the model sees a clean, unoccluded face, it may still search for the boundaries and textures it learned to expect. On the other hand, when the model encounters the exact occlusions it was trained on, its specialized attention aligns perfectly with the familiar structure, sometimes producing even more accurate predictions than on a fully visible face.

7 Thesis direction

Our findings require a refinement of our initial hypothesis and provide a clear roadmap for the next phase of this thesis. While our data confirms that target-driven data augmentation allows TokenHPE to adapt remarkably well to known occlusions (masks and glasses), the failure analysis revealed two critical limitations: scale sensitivity, where small faces collapse into too few tokens, and domain sensitivity, where out-of-distribution objects like headphones degrade performance. Therefore, our revised hypothesis would be that while semantic augmentation successfully teaches the model to handle specific accessories, achieving true real-world robustness requires an architecture that is invariant to both target scale and specific object textures.

To test this hypothesis, the next phase of our research could execute a concrete, two step plan:

- **Multi-Scale Tokenization:** To resolve the performance drop induced by the scale, we will modify the TokenHPE input pipeline to mix smaller patch sizes (like 8 x 8 pixels) with the standard 16 x 16 patches. This will artificially preserve token density for distant or small faces, providing the self-attention mechanism with the necessary spatial resolution to calculate the Pitch, Yaw and Roll matrices with precision.
- **Adversarial occlusion of the most informative parts:** Inspired by masked image models like ADIOS [11], in order to break the model’s dependence on specific shapes, we will implement a dynamic training layer that randomly "blinds" the most-attended image patches during backpropagation. Instead of relying on pre-rendered masks or glasses, TokenHPE will be forced to infer head orientation from arbitrary and fragmented anatomical data.

By implementing these two updates, we will transition TokenHPE from a model specialized in recognizing specific facial accessories to a truly generalized, scale-invariant and occlusion-agnostic head pose estimator.

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